AC3303 & AC3304

Introduction to Asset Pricing & Valuation

Autumn 2023 — Supplemental Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

Module	AC3303 & AC3304 — Introduction to Asset Pricing & Valuation
Session	Autumn 2023 (Supplemental)
Paper	1 of 1
External Examiner	Professor Thomas O'Connor
Internal Examiner	Dr. Joe Healy
Structure	Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)
This scheme covers	Question 1, Question 2 **and** Question 3 in full, so you can study either route through the paper

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme and is restated wherever it is especially relevant.

Quality check. Every calculation in this scheme has been independently verified and cross-checked for accuracy before being written up.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

Scenario

You are an investment analyst advising a client on shares in **Crock PLC** and **Cranny PLC**, both listed on the Stock Exchange. Three economic scenarios are given with associated probabilities and expected returns for each share:

State of the Economy	Probability	Return: Crock PLC	Return: Cranny PLC
Excellent Growth	0.30	12%	15%
No Growth at all	0.25	4%	-1%
Negative Growth	0.45	-8%	-7%
Total	**1.00**		

(a) Required

10 Marks

Calculate the Expected Return and Standard Deviation risk for both shares.

FULL MODEL ANSWER

The expected return of each share is the probability-weighted average of its returns across the three states. The standard deviation is the square root of the probability-weighted sum of squared deviations from that expected return.

State	Prob.	Return Crock	Prob × Return	Return Cranny	Prob × Return
Excellent Growth	0.30	12%	3.60	15%	4.50
No Growth at all	0.25	4%	1.00	-1%	-0.25
Negative Growth	0.45	-8%	-3.60	-7%	-3.15
E(R)			**1.00%**		**1.10%**

$E(R_{Crock})$ = 1.00% **$E(R_{Cranny})$ = 1.10%**

State	Prob.	(R−E(R)) Crock	Squared × Prob	(R−E(R)) Cranny	Squared × Prob
Excellent Growth	0.30	11.00	36.30	13.90	57.96
No Growth at all	0.25	3.00	2.25	-2.10	1.10
Negative Growth	0.45	-9.00	36.45	-8.10	29.52
Variance			**75.00**		**88.59**

Var(Crock) = 75.00 → SD(Crock) = 8.66%
Var(Cranny) = 88.59 → SD(Cranny) = 9.41%

Reconciliation check. Probabilities sum to exactly 1.00 and each squared-deviation column was recomputed independently from the raw state returns (not from a rounded intermediate E(R)), so no rounding drift enters the variance figures.

Mark Allocation — Part (a), 10 Marks

Tier	Marks	What separates this tier
9–10	9–10	Both expected returns and both standard deviations fully correct with clear probability-weighted working shown for each share.
6–8	6–8	Both expected returns correct; one standard deviation correct and the other has a minor arithmetic slip carried through consistently (own-figure).
3–5	3–5	Correct method (probability-weighted mean and variance) applied but with a computational error affecting more than one figure.
0–2	0–2	Wrong technique used (e.g. simple average instead of probability-weighted) or no workable attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(b) Required **5 Marks**

Calculate the co-variance between returns on shares in Crock Plc and on shares in Cranny Plc.

FULL MODEL ANSWER

Covariance is the probability-weighted sum of the product of each share's deviation from its own expected return.

State	Prob.	Dev. Crock	Dev. Cranny	Product	Prob × Product
Excellent Growth	0.30	11.00	13.90	152.90	45.87
No Growth at all	0.25	3.00	-2.10	-6.30	-1.575
Negative Growth	0.45	-9.00	-8.10	72.90	32.805
Covariance					**77.10**

Cov(Crock, Cranny) = 77.10

The strongly positive covariance means the two shares' returns tend to move together across economic states — both do well in growth and both fall in the negative-growth scenario. This is consistent with Crock and Cranny both being cyclical, share-price-sensitive stocks.

Mark Allocation — Part (b), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct covariance with full working showing deviations, products, and probability weighting.
3–4	3–4	Correct method but a minor own-figure carry-through error from part (a)'s expected returns.
1–2	1–2	Some correct elements (e.g. correct deviations) but incorrect combination or formula.
0	0	No credible attempt or fundamentally wrong approach (e.g. simple correlation instead of covariance).

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(c) Required **6 Marks**

Calculate the correlation coefficient between returns on shares in Crock Plc and on shares in Cranny Plc. Explain what this correlation coefficient means.

FULL MODEL ANSWER

The correlation coefficient standardises covariance by the product of the two standard deviations:

$$\rho = \text{Cov(Crock, Cranny)} \div (SD_{Crock} \times SD_{Cranny}) = 77.10 \div (8.66 \times 9.41) = 0.946$$

Strength and direction: A correlation coefficient of 0.946 is close to the maximum possible value of +1, indicating a very strong positive linear relationship between the two shares' returns.

What it means for diversification: Because the two shares move almost in lockstep, combining them in a portfolio provides very little diversification benefit — the portfolio's risk will sit close to the weighted average of the two individual standard deviations rather than being meaningfully reduced, since there is almost no offsetting movement between the assets.

Economic interpretation: The high correlation is consistent with both companies being exposed to the same broad economic cycle (their returns are driven by the same set of economic-state outcomes), which is typical of two companies operating in related, cyclically-sensitive sectors.

Mark Allocation — Part (c), 6 Marks

Tier	Marks	What separates this tier
6	6	Correct correlation coefficient (own-figure from parts (a) and (b)) plus a clear explanation covering both strength/direction and the diversification implication.
4–5	4–5	Correct coefficient with a partial explanation (e.g. states it is strongly positive but does not link to diversification).
2–3	2–3	Correct method but computational error, or correct coefficient with no meaningful explanation.
0–1	0–1	Incorrect formula (e.g. correlation not bounded between -1 and +1) or no attempt.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(d) Required

7 Marks

Determine the Expected Return and Standard Deviation risk of a portfolio where the investor decides to invest 35% of their money in Crock Plc and the remainder in Cranny Plc.

FULL MODEL ANSWER

Portfolio expected return is the weighted average of the two expected returns. Portfolio variance uses the full two-asset formula, incorporating the covariance calculated in part (b):

$$E(R_p) = w_1E(R_1) + w_2E(R_2)$$

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2Cov(1,2)$$

Component	Value
Weight in Crock (w_1)	0.35
Weight in Cranny (w_2)	0.65
$E(R_p)$ = 0.35(1.00%) + 0.65(1.10%)	**1.065%**
$w_1^2\sigma_1^2 = 0.35^2 \times 75.00$	9.1875
$w_2^2\sigma_2^2 = 0.65^2 \times 88.59$	37.4293
$2w_1w_2Cov = 2 \times 0.35 \times 0.65 \times 77.10$	35.0805
Portfolio Variance σ_p^2	**81.6973**
Portfolio Standard Deviation σ_p	**9.04%**

Reconciliation check. Because the correlation between the two shares is very high (0.946, from part (c)), the portfolio standard deviation of 9.04% sits almost exactly between the two individual standard deviations (8.66% and 9.41%) rather than below the lower of the two — this is the expected signature of a near-perfectly correlated pairing and confirms the covariance term has been applied correctly rather than ignored.

Mark Allocation — Part (d), 7 Marks

Tier	Marks	What separates this tier
6–7	6–7	Correct portfolio expected return and correct portfolio standard deviation using the full variance-covariance formula (not a simple weighted average of the two SDs).
4–5	4–5	Correct portfolio return; portfolio variance formula correct in structure but with an arithmetic slip, or covariance term omitted/misapplied (own-figure applied thereafter).
2–3	2–3	Portfolio return correct only, or variance calculated as a simple weighted average of variances with no covariance term.
0–1	0–1	No credible attempt at portfolio risk-return calculation.

Question 1 (Compulsory)

50 Marks — Portfolio Theory & Efficient Market Hypothesis

(e) Required **22 Marks**

Outline the implications of the Efficient Market Hypothesis for Companies and investors.

FULL MODEL ANSWER

The Efficient Market Hypothesis (EMH) holds that security prices fully and rapidly reflect all available information, so that no investor can consistently earn abnormal (risk-adjusted) returns using that information. The standard framework distinguishes three forms, and each carries distinct implications.

The three forms of market efficiency

Form	Information set reflected in price	Implication
Weak form	All past price and volume data	Technical analysis (chart patterns) cannot generate abnormal returns
Semi-strong form	All publicly available information (accounts, announcements, news)	Fundamental analysis of public information cannot generate abnormal returns
Strong form	All information, including private/insider information	Even insiders cannot consistently outperform the market

Implications for investors

Active management is difficult to justify: If markets are semi-strong form efficient, professional stock-picking and market-timing should not reliably beat a well-diversified market portfolio net of fees, which is the core argument for passive/index investing.

Diversification, not analysis, is the reliable lever: Since abnormal returns from information analysis are not sustainably available, investors are better served focusing on diversification to manage risk (as in parts (a)–(d) above) rather than trying to identify mispriced shares.

Prices react quickly to news: Investors should expect share prices to adjust rapidly — often within minutes — to new information, meaning there is little scope to trade profitably on information that has already been publicly released.

Technical trading rules add cost without benefit: Weak-form efficiency implies that strategies based purely on past price patterns are unlikely to outperform a simple buy-and-hold approach once transaction costs are considered.

Question 1 (continued)

Implications for companies (management)

> **The share price is a fair, unbiased signal:** In an efficient market, management can treat the current share price as an accurate reflection of the company's fundamental value given publicly available information, which is useful for evaluating whether the market has correctly priced a strategic decision.

> **No incentive to mislead the market with cosmetic reporting:** Because semi-strong efficiency implies the market can see through purely cosmetic accounting choices (e.g. changes in reporting presentation with no cash-flow effect), management gains nothing from earnings management that does not change underlying economics — the market should not be fooled for long.

> **Timing decisions around information releases matters less than substance:** Since prices react to the substance of news rather than its timing or framing, management should focus on making genuinely value-creating decisions (e.g. positive-NPV investments) rather than trying to time announcements to influence short-term price movements.

> **Financing decisions signal information to the market:** Because prices are highly sensitive to new information, a company's own financing choices (e.g. issuing new equity) can itself be interpreted as a signal — the EMH framework is central to understanding why announcements of share issues, buybacks, or dividend changes often move the price even without new external information.

Other acceptable answers. Candidates who structure this answer around the definition of EMH, the three forms, and a clearly separated discussion of investor implications (diversification/passive investing/rapid price adjustment) versus company implications (fair pricing signal, no benefit to cosmetic disclosure, financing/signalling effects) should receive full credit regardless of which specific implication or ordering they choose, provided the underlying economic reasoning is sound.

Mark Allocation — Part (e), 22 Marks

Tier	Marks	What separates this tier
18–22	18–22	Clear, accurate definition of EMH and all three forms; well-developed, distinct implications for both investors AND companies with sound economic reasoning throughout.
13–17	13–17	Good definition and forms covered; implications discussed for both investors and companies but less developed, or one side (investors/companies) noticeably thinner than the other.
8–12	8–12	Definition present but forms not fully or correctly distinguished; implications largely one-sided (e.g. investors only) or generic without linking back to EMH.
0–7	0–7	Definition of EMH absent or materially incorrect; no meaningful discussion of implications.

Question 1 Total: 50 Marks

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

How to use this scheme. The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally toward the paper. This scheme covers both in full so you can revise whichever route you are more comfortable with, or compare your strengths on both before the exam.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(a) Required

8 Marks

Calculate the numbers a)–e) to complete the following table (Present Value, Years, Interest Rate, Future Value):

FULL MODEL ANSWER

Every row uses the single-sum time-value-of-money relationship $FV = PV(1+r)^n$, rearranged for whichever variable is missing.

Present Value	Years	Interest Rate	Future Value
900	1	8%	a) ?
900	19	3%	b) ?
c) ?	4	7%	300,000
8,000	5	d) ?	10,000
7,000	40	e) ?	49,000

Row	Formula used	Answer
a)	$FV = 900 \times (1.08)^1$	**972.00**
b)	$FV = 900 \times (1.03)^1$&sup9;	**1,578.16**
c)	$PV = 300{,}000 \div (1.07)^4$	**228,868.56**
d)	$r = (10{,}000/8{,}000)^{1/5} - 1$	**4.56%**
e)	$r = (49{,}000/7{,}000)^{1/40} - 1$	**4.99%**

Each answer was independently verified in Python using the same $FV = PV(1+r)^n$ relationship rearranged algebraically — no value was estimated or interpolated.

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7–8	7–8	All five values (a–e) correct, including both rate-solving rows (d and e) correctly rearranged.
5–6	5–6	Three or four values correct; typically the FV/PV rows correct with one rate-solving row (d or e) wrong or omitted.
2–4	2–4	One or two values correct; correct general formula recognised but frequent rearrangement errors.
0–1	0–1	Formula not recognised or applied incorrectly throughout.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(b) Required **12 Marks**

An 8-year U.S. Treasury bond with a face value of $1,000 pays an annual coupon of 7.50%. The reported yield to maturity is 2%.

a. What is the current price of the bond? (4 Marks)

b. If the yield to maturity changes to 1%, what would be the price? (4 Marks)

c. If the yield to maturity changes to 4%, what would be the price? (4 Marks)

FULL MODEL ANSWER

Bond price = PV of the coupon annuity + PV of the face value repaid at maturity. Because the bond pays an **annual** coupon, both legs are discounted annually at the YTM (not semi-annually):

$$\text{Price} = C \times [1 - (1+r)^{-n}] / r \; + \; FV / (1+r)^{n}$$

Annual coupon C = 7.50% × $1,000 = **$75.00**, n = 8 years.

Yield to Maturity	PV of Coupons	PV of Face Value ($1,000)	Bond Price
2.0% (given)	549.41	853.49	**$1,402.90**
1.0%	573.88	923.48	**$1,497.36**
4.0%	504.96	730.69	**$1,235.65**

Sense check. The bond's coupon (7.50%) is well above every YTM tested, so the bond should trade at a premium to face value in all three cases — the three prices above are all comfortably above $1,000, confirming the direction is correct. The inverse relationship between YTM and price also holds throughout: price falls as YTM rises from 1% → 2% → 4%.

Mark Allocation — Part (b), 12 Marks (4 per YTM scenario)

Tier	Marks	What separates this tier
4 (per scenario)	4	Correct bond price with both the coupon annuity PV and face value PV shown separately and summed correctly.
2-3 (per scenario)	2–3	Correct formula structure but an arithmetic error in one component (own-figure carried through).
1 (per scenario)	1	Coupon annuity or face value PV calculated correctly in isolation but not combined into a bond price.
0 (per scenario)	0	Wrong formula (e.g. simple interest, or semi-annual compounding applied despite the annual coupon) or no attempt.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(c) Required **20 Marks**

There are a number of practical implications and problems in terms of the usefulness of the Capital Asset Pricing Model that need to be understood before applying the model. Discuss these implications and problems and how they impact on the applicability of the model itself.

FULL MODEL ANSWER

The CAPM is the standard tool for estimating a required/expected return given systematic risk, but its practical application faces several well-documented problems:

Estimating beta is backward-looking and unstable: Beta is typically estimated from historical regression of a share's returns against a market index, but a company's operations, capital structure and risk profile change over time, so a historical beta may be a poor estimate of the beta that will actually prevail going forward.

Choice of market proxy: The CAPM requires the return on 'the market portfolio' — theoretically every risky asset in the economy — but in practice a proxy such as a stock index (e.g. S&P 500) is used, which is not the true market portfolio (it excludes bonds, property, human capital, and non-listed assets), introducing potential mismeasurement (Roll's critique).

Single-factor model — empirical evidence of missing risk factors: Extensive empirical research (e.g. Fama and French) finds that factors beyond market beta — notably firm size and the book-to-market ratio — help explain the cross-section of stock returns, suggesting CAPM's single beta does not fully capture priced risk, which has led to multi-factor alternatives (e.g. the Fama-French three-factor model).

Risk-free rate and equity risk premium are themselves estimates: The risk-free rate (usually a government bond yield) and the market risk premium both require judgement calls — the risk premium in particular can vary substantially depending on the historical period and method used to estimate it, feeding directly into the required-return output.

Assumes a single-period, frictionless, well-diversified world: CAPM assumes investors can borrow and lend freely at the risk-free rate, face no taxes or transaction costs, and hold well-diversified portfolios — assumptions that do not hold precisely in practice, particularly the assumption of unrestricted risk-free borrowing.

Practical impact on applicability: Despite these problems, CAPM remains widely used in practice (e.g. for cost of equity in capital budgeting) because it is simple, transparent, and its inputs are observable and auditable — but practitioners should treat the output as an estimate with a margin of error rather than a precise figure, and where possible cross-check it against alternative approaches (e.g. dividend growth model, multi-factor models).

Other acceptable answers. Candidates may instead or additionally discuss the assumption of homogeneous expectations, the single-period nature of the model, or specific named empirical studies (e.g. Fama-French 1992/1993, Roll 1977). Any well-explained subset of the standard critique list, correctly linked to practical applicability, should receive full credit — there is no single "correct" list of points for a question of this type.

Mark Allocation — Part (c), 20 Marks

Tier	Marks	What separates this tier
16–20	16–20	At least four to five distinct, well-explained problems (e.g. beta instability, market proxy, missing risk factors, input estimation, assumptions) each clearly linked to practical impact on applicability.
11–15	11–15	Three to four relevant problems identified with reasonable explanation, but linkage to practical applicability underdeveloped.
6–10	6–10	One to two problems identified, or points listed without explanation of why they matter for using the model in practice.
0–5	0–5	No accurate discussion of CAPM's limitations, or answer confuses CAPM with an unrelated model.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(d) Required **10 Marks**

In the general context of Corporate Finance, discuss the forms that the Financing Decision can take for an organisation.

FULL MODEL ANSWER

The financing decision concerns how a company raises the funds needed to support its investment decisions, and centres on the choice and mix of sources of finance:

Equity finance: Raising funds by issuing ordinary shares (e.g. via an IPO, rights issue, or placing), which gives shareholders a residual claim on profits and voting rights but carries no obligation to pay a fixed return, making it the most flexible but also typically the most expensive source of capital.

Debt finance: Borrowing via bank loans, bonds, or other fixed-income instruments, which carries a contractual obligation to pay interest and repay principal regardless of profitability, but is generally cheaper than equity due to its lower risk to the provider and the tax deductibility of interest.

Retained earnings (internal finance): Using internally generated cash flow rather than external capital markets, which avoids issuance costs and dilution but is limited by the scale of the company's own profitability and cash generation.

Hybrid instruments: Instruments such as convertible bonds or preference shares that combine features of both debt and equity, allowing companies to tailor the risk/return and control characteristics of the capital they raise.

The capital structure decision: Beyond choosing individual instruments, management must decide the overall mix (the debt-to-equity ratio) that minimises the firm's weighted average cost of capital while keeping financial risk (the risk of not being able to meet fixed obligations) at an acceptable level.

Short-term versus long-term financing: The financing decision also spans the maturity structure — short-term financing (e.g. trade credit, overdrafts) for working capital needs versus long-term financing (e.g. term loans, bonds, equity) for fixed asset investment — with the matching principle generally favouring financing assets with liabilities of a similar maturity.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8–10	8–10	Clear coverage of equity, debt and retained earnings as core financing forms, with sound discussion of the capital structure trade-off and/or maturity matching.
5–7	5–7	Equity and debt covered adequately but retained earnings, hybrids, or the capital structure trade-off missing or underdeveloped.
2–4	2–4	Only one financing form discussed in any depth, or discussion is generic without reference to the trade-offs between sources.
0–1	0–1	No meaningful discussion of financing forms.

Question 2 Total: 50 Marks

Question 3

50 Marks — Choice question: alternative to Question 2

(a) Required

8 Marks

Using the Capital Asset Pricing Model (CAPM) calculate the expected return for Share Z, assuming a beta factor of 0.60, a risk free rate of interest of 3%, and a market risk premium of 8%.

FULL MODEL ANSWER

$$E(R) = R_f + \beta(\text{Market Risk Premium})$$

Component	Value
Risk-free rate, R_f	3.00%
Beta, β	0.60
Market risk premium	8.00%
β × Market risk premium = 0.60 × 8%	4.80%
E(R) = 3% + 4.80%	**7.80%**

Interpretation: Because Share Z has a beta below 1 (0.60), it is less volatile than the market portfolio, so its required return of 7.80% sits below the assumed market return of 11% (3% + 8%) — consistent with it carrying below-average systematic risk.

Mark Allocation — Part (a), 8 Marks

Tier	Marks	What separates this tier
7–8	7–8	Correct CAPM formula correctly applied with correct final expected return, and a sound interpretation of beta.
4–6	4–6	Correct formula and calculation but no interpretation of the beta figure, or a minor arithmetic slip.
1–3	1–3	Formula recalled but misapplied (e.g. beta and premium not multiplied together first).
0	0	No credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(b) Required

12 Marks

Hatchett Plc earnings and dividends per share are expected to grow by 4% per annum. This growth will stop after 3 years, in year 4 growth will slow to a constant rate of 1% thereafter. Assume that this year's dividend is $3.00 per share and the market capitalisation rate is 6%. This year's EPS is $10 per share.
a) What is Hatchett Plc's current stock price? (8 Marks)
b) Calculate the P/E Ratio and briefly explain what you conclude from the relationship between the P/E ratio and the growth prospects for Hatchett Plc. (4 Marks)

FULL MODEL ANSWER

This is a two-stage dividend discount model: dividends grow at 4% for three years, then settle into a constant 1% growth perpetuity from year 4 onward. D_0 = $3.00 is this year's dividend just paid.

Year	Growth applied	Dividend	PV factor @ 6%	Present Value
1	$D_0 \times 1.04$	$3.1200	0.9434	$2.9434
2	$D_1 \times 1.04$	$3.2448	0.8900	$2.8879
3	$D_2 \times 1.04$	$3.3758	0.8396	$2.8334
4 onward	$D_3 \times 1.01$, growing at 1% forever	D_4 = $3.4096	—	—

Terminal value at end of Year 3 (value of the year-4-onward growing perpetuity, discounted back to t=3):

$$\mathbf{TV_3 = D_4 \div (r - g) = 3.4096 \div (0.06 - 0.01) = \$68.19}$$

PV of terminal value = $68.19 ÷ $(1.06)^3$ = **$57.23**

Component	Present Value
PV of Year 1 dividend	$2.94
PV of Year 2 dividend	$2.89
PV of Year 3 dividend	$2.83
PV of terminal value	$57.23
Current Stock Price	**$65.90**

A small rounding difference (a few cents) versus a hand-calculated answer is acceptable and is an own-figure matter, provided the two-stage structure (explicit dividends for years 1–3, then a growing perpetuity from year 4 discounted back three years) is applied correctly.

Question 3 (continued)

Part b) P/E Ratio

Component	Value
Current stock price	$65.90
This year's EPS	$10.00
P/E ratio = Price ÷ EPS	**6.59**

Interpretation: A P/E ratio of 6.59 is relatively modest. Because a large part of the standard PVGO decomposition of P/E reflects expected future growth, a P/E in this range signals that the market is pricing in fairly limited growth prospects for Hatchett — consistent with the 4% growth phase being short-lived (only three years) before settling to a low 1% long-run rate, which caps the value of future growth opportunities relative to the company's current earnings power.

Mark Allocation — Part (b), 12 Marks

Tier	Marks	What separates this tier
10–12	10–12	Correct two-stage DDM structure with all three explicit dividends and correctly-discounted terminal value; correct price, correct P/E, and a sound explanation linking P/E to growth prospects.
7–9	7–9	Correct structure with a minor arithmetic slip in one component (own-figure), or price fully correct but P/E interpretation thin.
4–6	4–6	Growth stages confused (e.g. terminal value discounted back the wrong number of years, or D0 treated as D1), but method otherwise recognisable as a DDM approach.
0–3	0–3	Single-stage (Gordon growth) model used throughout with no attempt to separate the two growth phases, or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(c)(i) Required

8 Marks

An investor is worried that the price of shares they currently hold in Bright PLC are currently overpriced at €35.00 and that they may well fall in price in the coming weeks. However, the investor does not want to sell the shares, fearing they are wrong and the share price might actually rise. The premium on a three-month Bright PLC Put option with an exercise price of €35.00 is currently €3.00.

a) Given the information provided, indicate the option strategy the investor would employ to protect themselves against any share price decrease.

b) Calculate the profit/(loss) that would accrue from such a strategy if Bright PLC's share price after three months was: (a) €40 (b) €30. Supplement your answer with appropriate payoff diagrams.

FULL MODEL ANSWER

Strategy: The investor should buy a put option on Bright PLC with a €35.00 exercise price (a **protective put**). This gives the investor the right, but not the obligation, to sell their shares at €35.00, locking in a floor value while keeping full upside if the share price rises instead — exactly matching the investor's wish to hedge downside risk without selling the shares outright.

Profit/loss on the put option position itself, at expiration, per share:

Profit = Max(Exercise Price − Share Price, 0) − Premium Paid

Share Price at Expiry	Put exercised?	Intrinsic Value	Less: Premium Paid	Profit/(Loss) on Put
€40.00	No — price above strike	€0.00	€3.00	**(€3.00)**
€30.00	Yes — sell at €35 vs market €30	€5.00	€3.00	**€2.00**

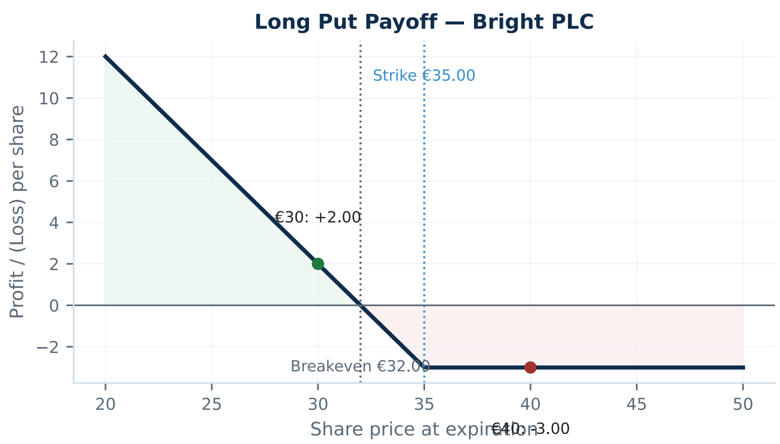

Fig. 1 — Long put payoff, Bright PLC (strike €35.00, premium €3.00, breakeven €32.00)

Scope of this profit/loss figure. The €30 outcome shows a €2.00 gain and the €40 outcome shows a €3.00 loss *on the put option itself*. The investor's overall wealth also includes the change in value of the shares already held, but since the paper does not give the original purchase price of those shares, that combined figure cannot be calculated — the put's own profit/loss (shown here) is what is being asked for, consistent with how this option-strategy question type has been marked in other papers in this series. At €30, the put's €2.00 gain partially offsets the €5.00 fall in the shares (from €35 to €30) — this is the hedge working as intended, not full protection, because the €3.00 premium is a sunk cost of insurance.

Mark Allocation — Part (c)(i), 8 Marks

Tier	Marks	What separates this tier
7–8	7–8	Correct strategy identified (buy a put / protective put) with clear reasoning; both profit/loss figures correct; payoff diagram correctly shaped (kinked at strike, breakeven at €32).
5–6	5–6	Correct strategy and both figures correct, but diagram missing or materially incorrect (e.g. wrong kink point).
2–4	2–4	Correct strategy identified but one or both profit/loss figures wrong, or a call option used instead of a put.
0–1	0–1	Wrong strategy (e.g. selling the shares, or writing an option) or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(c)(ii) Required

12 Marks

"The Black-Scholes option pricing formula puts forward five variables that affect the traded price of an option contract". Discuss this statement, describing the five relevant variables and how each affects option prices.

FULL MODEL ANSWER

The Black-Scholes model identifies five inputs that determine an option's theoretical price:

Variable	Effect on Call price	Effect on Put price
Current share price (S)	Increases as S rises	Decreases as S rises
Exercise/strike price (K)	Decreases as K rises	Increases as K rises
Time to expiration (T)	Increases with more time (more chance to move in-the-money)	Generally increases with more time
Volatility of the underlying (σ)	Increases with volatility	Increases with volatility
Risk-free interest rate (r)	Increases as r rises	Decreases as r rises

Share price and strike price: A call becomes more valuable the further the share price is above the strike (more intrinsic value), while a put becomes more valuable the further the share price is below the strike — this is the direct in-the-money effect for each option type.

Time to expiration: More time means more opportunity for the underlying price to move favourably before expiry, so both calls and puts on non-dividend-paying shares are generally worth more with more time remaining (time value).

Volatility: Because option payoffs are asymmetric (losses capped at the premium paid, but upside is uncapped for the holder), higher volatility increases the probability of a large favourable move without symmetrically increasing the cost of a large unfavourable move — so both call and put values rise with volatility.

Risk-free interest rate: A higher risk-free rate raises the cost of carrying the underlying and lowers the present value of the exercise price paid in the future, which increases call values (since the strike is effectively 'cheaper' in present-value terms) and decreases put values.

Candidates who correctly identify all five variables and their qualitative direction of effect on both call and put prices should receive full credit — the exact numerical sensitivities (the option 'Greeks': Delta, Vega, Theta, Rho) are not required at this level, only the direction of the relationship.

Mark Allocation — Part (c)(ii), 12 Marks

Tier	Marks	What separates this tier
10–12	10–12	All five variables correctly identified with accurate direction of effect on BOTH calls and puts, and sound underlying reasoning.
7–9	7–9	Four or five variables identified with mostly correct direction of effect, minor errors on one variable (e.g. interest rate effect on puts).
4–6	4–6	Two to three variables correctly identified, or direction of effect frequently confused between calls and puts.
0–3	0–3	Fewer than two variables correctly identified, or fundamental misunderstanding of what the variables represent.

Question 3

50 Marks — Choice question: alternative to Question 2

(d) Required **10 Marks**

Describe the main stakeholders in an organisation.

FULL MODEL ANSWER

Stakeholders are any group with an interest in, or affected by, the decisions and performance of an organisation. The main groups are:

Stakeholder	Primary interest
Shareholders / owners	Return on investment: dividends and capital growth; ultimate residual claimants on profit
Management / employees	Remuneration, job security, career progression; day-to-day decision-making authority
Debt holders / creditors	Timely payment of interest and repayment of principal; protection of the value of their claim
Customers	Quality, price and reliability of the goods or services provided
Suppliers	Timely payment and a stable, ongoing trading relationship
Government / regulators	Tax revenue, compliance with law and regulation, wider economic and social impact
Local community / society	Employment, environmental impact, and the broader social footprint of the organisation

Why the distinction matters: Shareholder wealth maximisation is the traditional objective of corporate finance, but management must balance shareholders' interests against those of other stakeholders — for example, debt holders' interest in limiting risk-taking (addressed through debt covenants), or employees' and communities' interest in the sustainability of the business — because sustained conflict with any one group can ultimately undermine shareholder value itself.

Link to agency theory: The stakeholder framework overlaps with agency theory: management (the agent) may not always act purely in shareholders' (the principal's) interest, and equally shareholders' interests may not always align with the interests of other stakeholder groups, which is why governance mechanisms exist to balance these competing claims.

Mark Allocation — Part (d), 10 Marks

Tier	Marks	What separates this tier
8–10	8–10	At least five distinct stakeholder groups identified with their primary interest correctly described, plus discussion of why balancing these interests matters.
5–7	5–7	Three to four stakeholder groups correctly identified and described, but no discussion of why the distinction matters or how interests can conflict.
2–4	2–4	One to two stakeholder groups identified, or groups listed without describing their interest.
0–1	0–1	No meaningful list of stakeholders.

Question 3 Total: 50 Marks